Secure Web-Based Banking System: UI/UX Design Documentation

Course: SCMP 401 – Software Engineering Studio **Milestone:** 1 **Prepared by:** Akar Omer – UI/UX Designer

Team Member	Assigned Role
Shakar Tahsin	Product Owner / Project Owner
Shadyar Faruq	Backend Developer
Akar Walid	Database Designer
Akar Omer	UI/UX Designer
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This document translates the Product Owner's vision, five user requirements (UR-1 to UR-5) and five user stories (US-01 to US-05) into a user-interface and user-experience design: users, flows, screens, components, messages, visual style, accessibility and usability testing.

1. Design Purpose and Goals

The interface must let a bank customer log in securely, create an account, check a balance, deposit and withdraw, without confusion or risk of costly mistakes.

Goal	What it means for the design
Trust and security	Clear, calm visuals. Security steps (MFA, lockout) are explained, not hidden.
Clarity	One main task per screen. Balances and amounts are always the most prominent element.
Error prevention	Inline validation, confirmation before money moves, and rules shown before the user submits.

Goal	What it means for the design
Efficiency	Any core task is reachable in three clicks or fewer after login.
Accessibility	Usable by keyboard, screen reader, and on phone, tablet and desktop.
Consistency	Same components, wording and layout across all screens.

2. Users and Context

Primary user: Bank Customer. Uses the web application to log in and manage their accounts (Product Vision, Section 3).

Persona 1 – Dana, 24, university student (Savings focus)

- Wants to deposit small amounts and see her balance quickly.
- Pain point: unsure why a withdrawal is refused. The UI must explain the $100 minimum clearly.

Persona 2 – Karwan, 41, shop owner (Current Account focus)

- Makes frequent deposits and withdrawals, sometimes using the overdraft.
- Pain point: needs to know exactly how much he can still withdraw, including the $500 overdraft.

Persona 3 – Hana, 58, occasional online banking user

- Needs large text, simple wording and few steps.
- Pain point: anxious about security steps and about making mistakes with money.

Personas are design assumptions made to guide decisions; they can be replaced with real feedback from usability testing.

3. Traceability: Requirements to Design

Requirement	User Story	Screens	Key UX concern
UR-1 Authentication	US-01	S1 Login, S2 MFA, S3 Locked state	Explain MFA and lockout without revealing which credential was wrong
UR-2 Account Creation	US-02	S5 Create Account	Make the difference between Current and Savings obvious; validate inputs
UR-3 Balance Inquiry	US-03	S4 Dashboard, S6 Account Details	Balance visible fast (target < 1.5 s); show a loading state
UR-4 Deposit	US-04	S7 Deposit, S9 Result	Only positive amounts; clear success record
UR-5 Withdraw	US-05	S8 Withdraw, S9 Result	Show overdraft and minimum-balance rules; reject with a clear reason

4. Information Architecture

```
Login (S1)
└── MFA Verification (S2)
    └── Dashboard / My Accounts (S4)
        ├── Create Account (S5)
        └── Account Details (S6)
            ├── Balance and Recent Transactions
            ├── Deposit (S7)
            └── Withdraw (S8)
                └── Transaction Result (S9)
Sign out (always available in header after login)
```

Navigation model: a simple top bar with the bank name, the customer's name and a *Sign out* button. The dashboard is the home hub; every other screen has a visible *Back to accounts* link. No deep menus, because the scope has only five functions.

5. Main User Flows

Flow A – Login with MFA (US-01)

1. Customer opens the site and sees the Login screen.
2. Enters user ID and password and presses **Log in**.
3. If valid, the MFA screen asks for the second factor.
4. If MFA is correct, the customer lands on the Dashboard.
5. If credentials are wrong, an error appears. After three consecutive failures within five minutes, the Locked screen is shown for 15 minutes.
6. If someone tries to open a protected page without authenticating, they are sent back to Login.

Flow B – Create an account (US-02)

Dashboard → **Create new account** → choose Current or Savings → fill required fields → review → **Create account** → success message → new account appears on the Dashboard.

Flow C – View balance (US-03)

Dashboard → account card shows balance, or tap the card to open Account Details. Only the customer's own active accounts are listed.

Flow D – Deposit (US-04)

Account Details → **Deposit** → enter amount → **Review** → **Confirm deposit** → Result screen with updated balance and transaction record.

Flow E – Withdraw (US-05)

Account Details → **Withdraw** → enter amount → rules check shown live → **Review** → **Confirm withdrawal** → Result screen (success or clear rejection reason).

6. Screen Specifications

Each screen is described as a wireframe in words so developers can build it and the team can review it before high-fidelity mockups.

S1 – Login

- **Layout:** centered card on a plain background, bank logo above.
- **Elements:** User ID field, Password field with show/hide toggle, **Log in** primary button, short security note ("Your session is protected with multi-factor authentication").
- **States:** default, field focus, validation error, loading (button disabled with spinner), error after failed login.
- **Rules:** the error message must not say which of user ID or password was wrong.
- **Supports:** UR-1, US-01.

S2 – MFA Verification

- **Elements:** explanation text ("Enter the verification code to finish logging in"), code input, **Verify** button, **Back to login** link.
- **States:** default, invalid code, loading.
- **Note:** the exact second factor (SMS, email, authenticator app) is to be confirmed with the Architecture & Security member. The screen is designed around a single code input so it works with any of these.

S3 – Account Locked

- **Trigger:** third consecutive failed login within five minutes.
- **Elements:** lock icon, message "Your account is temporarily locked for 15 minutes for your security," and a **Back to login** link. A countdown timer is optional and depends on what the backend can supply.
- **Rule:** the locked state must also appear if the user returns during the 15 minutes.

S4 – Dashboard / My Accounts

- **Elements:** greeting, list of **account cards** (account type, masked account number, current balance), **Create new account** button.
- **Empty state:** "You don't have an account yet. Create your first account." with a primary button.
- **States:** loading (skeleton cards), loaded, error with **Try again**.
- **Supports:** UR-3, US-03.

S5 – Create Account

- **Elements:** two large selectable option cards, *Current Account* and *Savings Account*, each with a short summary of its rules:
  - Current: for frequent transactions, overdraft up to $500.
  - Savings: earns 2.5% annual interest, $100 minimum balance for withdrawals.
- Required-information fields (to be finalised with the Database Designer), **Create account** and **Cancel** buttons.
- **Validation:** inline messages next to the field, e.g. "This field is required." Nothing is created until all required data is valid.
- **Supports:** UR-2, US-02.

S6 – Account Details

- **Elements:** account type and number, large **current balance**, buttons **Deposit** and **Withdraw**, recent transactions list (date, type, amount).
- **Savings accounts** also show a small note: "Minimum balance for withdrawals: $100." **Current accounts** show: "Overdraft available: up to $500."
- **Access rule:** if an account is not active or not owned by the customer, no balance is displayed, only a neutral message.

S7 – Deposit

- **Elements:** account name, amount field (numeric, currency symbol $), **Review** button, **Cancel**.
- **Validation:** amount must be greater than zero. Zero, negative or non-numeric values show an inline error and disable the button.
- **Confirmation step:** "You are depositing $X into [Account]. Confirm?" with **Confirm deposit** and **Edit**.

S8 – Withdraw

- **Elements:** account name, amount field, a **rules hint** under the field, **Review** button, **Cancel**.
- **Rules hint:**
  - Current: "You can withdraw up to your balance plus the $500 overdraft."
  - Savings: "Your balance must stay at or above $100 after withdrawal."
- **Validation:** amount greater than zero; live warning if the amount would break the rule.

- **Confirmation step:** shows the amount and the resulting balance before the user confirms.

S9 – Transaction Result

- **Success:** check icon, "Deposit successful" or "Withdrawal successful," amount, new balance, date and time, **Back to account** button.
- **Failure:** warning icon, clear reason (see Section 8), and the balance stated as unchanged, with **Try again** and **Back to account**.
- **Note:** the UI must never show a success message unless the backend confirms that both the balance update and the transaction record were completed together.

7. Component Library

Component	Variants / states
Primary button	default, hover, focus, pressed, disabled, loading
Secondary button / link	default, hover, focus, disabled
Text input	default, focus, filled, error, disabled
Password input	as text input plus show/hide toggle
Amount input	numeric keypad on mobile, $ prefix, error state
Account card	default, hover/focus, loading skeleton
Selectable option card	unselected, selected, focus
Alert / banner	success, error, warning, info
Confirmation panel	summary plus Confirm and Edit
Transactions list row	deposit, withdrawal
Top bar	logged-out (logo only), logged-in (name, Sign out)

8. Messages and Microcopy

Plain wording, no technical jargon, always say what happened and what to do next.

Situation	Message
Failed login	"The user ID or password is incorrect. Please try again."
Third failed login	"Your account is locked for 15 minutes for your security."
Invalid MFA code	"That code is not correct. Please check it and try again."
Empty required field	"This field is required."
Invalid amount	"Enter an amount greater than $0."
Savings minimum rule violated	"This withdrawal would take your balance below the $100 minimum."
Overdraft limit exceeded	"This amount is more than your balance plus the $500 overdraft limit."
Inactive or unavailable account	"This account is not available for transactions."
Operation failed	"Something went wrong and no changes were made. Your balance is unchanged."
Session expired	"You have been signed out. Please log in again."

9. Visual Design Guidelines

These are the proposed starting point; they can be adjusted after team review.

Colour palette

Role	Colour	Hex
Primary (trust, actions)	Deep blue	#1F3A5F
Primary hover	Darker blue	#162B47
Success	Green	#1E7F4F
Error	Red	#B3261E
Warning	Amber	#8A5A00

Role	Colour	Hex
Text	Near black	#1A1A1A
Secondary text	Grey	#5C5C5C
Background	Off white	#F6F8FA
Surface (cards)	White	#FFFFFF

Typography: a clean sans-serif (e.g. Inter or system UI font). Body 16 px, headings 24 and 32 px, balance figure 32 px bold. Line height 1.5.

Spacing and layout: 8 px spacing grid, rounded corners 8 px, one main column up to 960 px wide on desktop.

Iconography: simple outline icons with text labels; icons never carry meaning alone.

Colour rule: success and error are never shown by colour alone; they also use an icon and text.

10. Responsive Behaviour

Screen width	Behaviour
Mobile (< 600 px)	Single column, full-width buttons, account cards stacked, numeric keypad for amounts
Tablet (600–1024 px)	Single column with wider margins, cards in two columns
Desktop (> 1024 px)	Centered content, cards in grid, transaction list as table

11. Accessibility

Target: WCAG 2.1 level AA.

- Text contrast at least 4.5:1 against its background.
- Everything usable by keyboard, with a visible focus outline and logical tab order.
- Every input has a visible label (not placeholder only) and error messages linked to the field for screen readers.

- Status changes (success, error, lockout) announced to screen readers.
- Touch targets at least 44 × 44 px.
- Text can be enlarged to 200% without losing content.
- Session timeout warning before sign-out, if the backend supports it.

12. Security-Related UX Decisions

The Architecture & Security member owns the security design. The UI supports it as follows:

- Password field is masked by default, with an optional show toggle; the browser may offer a password manager.
- Login errors are generic so attackers cannot tell which credential was wrong.
- Account numbers are masked on cards (e.g. •••• 1234) and full only where needed.
- Sensitive data is never placed in the page URL.
- Money-moving actions require an explicit confirmation step.
- Buttons are disabled while a request is in progress to prevent double submission of deposits or withdrawals.
- A visible **Sign out** button on every logged-in page.
- The UI only communicates over the secure connection (TLS 1.3) defined by the team's security requirements.

13. Performance Considerations in the UI

- The balance inquiry target is under 1.5 seconds with 50 concurrent requests. The interface should show skeleton loaders immediately so waiting feels short, and avoid loading large images or heavy scripts on banking pages.
- Show a clear error with **Try again** if a request fails or times out.

14. Usability Testing Plan

Task	Success measure
Log in and complete MFA	Completed without help
Create a Savings Account	Completed in under 2 minutes

Task	Success measure
Find the balance of an account	Found in under 10 seconds
Deposit $50	Completed without error
Try to withdraw an amount that breaks the Savings rule	User understands why it failed
Recover from a wrong password	User understands what happened and what to do

Method: 5 classmates or friends complete the tasks on the prototype while the designer observes. Record completion, time, errors and comments, then list the top issues and fix them in the next iteration.

15. Design Tools and Deliverables

- **Wireframes and prototype:** Figma (low fidelity first, then high fidelity), with clickable flows for A to E.
- **Design handoff:** component specs, colours and spacing shared with the Backend Developer through the Figma file and this document.
- **Milestone 1 deliverables:** this documentation, wireframes for S1–S9, and the style guide in Section 9.

16. Dependencies and Open Questions

Item	Who to ask
Which MFA method will be used (code by SMS, email or app)?	Awder Muhamed
Which fields are required to create an account?	Akar Walid, Shakar Tahsin
Can the backend return remaining lockout time for a countdown?	Shadyar Faruq
Are there transaction-frequency limits on Savings accounts? The brief mentions them without detail, so the UI may need to display them.	Shakar Tahsin

Item	Who to ask
Are there any deposit rules beyond amount greater than zero? The brief says to confirm with the instructor.	Shakar Tahsin
Should interest earned (2.5%) be shown to the customer in this release?	Shakar Tahsin
Is a session timeout planned?	Awder Muhamed

17. UI Definition of Done

In line with the Product Owner's Definition of Done, a UI item is complete when:

- The screen matches its user story and acceptance criteria.
- All states (default, loading, error, empty, success) are designed.
- Invalid input and failure cases show the agreed messages.
- The design meets the accessibility checklist in Section 11.
- The design is reviewed by the Product Owner and shared with the developers.
- Changes to scope or business rules are discussed with the Product Owner and recorded in the main SRS.